

CO 1 ASSESSMENT TOOL- 3

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COURSE CODE: DSA0304

COURSE NAME: NATURAL LANGUAGE PROCESSOR

1.Intelligent Email and Password Validator

CODE:

```
import re
email = input("Enter Email: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")
email_pattern = r"^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$"
password_pattern = r"^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!]).{8,}$"
mobile_pattern = r"^[6-9]\d{9}$"
print("\n--- Validation Result ---")
if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
if re.fullmatch(mobile_pattern, mobile):
    print("Valid Mobile Number")
else:
    print("Invalid Mobile Number")
```

```
import re
email = input("Enter Email: ")
password = input("Enter Password: ")
mobile = input("Enter Mobile Number: ")
email_pattern = r"^[A-Za-z][A-Za-z0-9._]*@[A-Za-z]+\.(com|org|edu|net|in)$"
password_pattern = r"^(?=.*[A-Z])(?=.*[a-z])(?=.*\d)(?=.*[@#%&!]).{8,}$"
mobile_pattern = r"^[6-9]\d{9}$"
print("\n--- Validation Result ---")
if re.fullmatch(email_pattern, email):
    print("Valid Email")
else:
    print("Invalid Email")
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
if re.fullmatch(mobile_pattern, mobile):
    print("Valid Mobile Number")
else:
    print("Invalid Mobile Number")
```

```
Enter Email: 192424408.simats@saveetha.com
Enter Password: 123
Enter Mobile Number: 7989536452
```

```
--- Validation Result ---
Invalid Email
Weak Password
Valid Mobile Number
```

2.DFA Simulator

CODE:

```
states = ["q0", "q1", "q2"]
alphabet = ["a", "b"]
start = "q0"
final = "q2"
transition = {
    "q0": {"a": "q1", "b": "q0"},
    "q1": {"a": "q1", "b": "q2"},
    "q2": {"a": "q1", "b": "q0"}
}
strings = ["abaab", "aab", "abb", "abab"]
for string in strings:
    state = start
    path = [state]
    for symbol in string:
        state = transition[state][symbol]
        path.append(state)
    print("\nInput:", string)
    print("Transition Path:", " → ".join(path))
    if state in final:
        print("Accepted")
    else:
        print("Rejected")
```

```
states = ["q0", "q1", "q2"]
alphabet = ["a", "b"]
start = "q0"
final = "q2"
transition = {
    "q0": {"a": "q1", "b": "q0"},
    "q1": {"a": "q1", "b": "q2"},
    "q2": {"a": "q1", "b": "q0"}
}
strings = ["abaab", "aab", "abb", "abab"]
for string in strings:
    state = start
    path = [state]
    for symbol in string:
        state = transition[state][symbol]
        path.append(state)
    print("\nInput:", string)
    print("Transition Path:", " → ".join(path))
    if state in final:
        print("Accepted")
    else:
        print("Rejected")
```

```
Input: abaab
Transition Path: q0 → q1 → q2 → q1 → q1 → q2
Accepted
```

```
Input: aab
Transition Path: q0 → q1 → q1 → q2
Accepted
```

```
Input: abb
Transition Path: q0 → q1 → q2 → q0
Rejected
```

```
Input: abab
Transition Path: q0 → q1 → q2 → q1 → q2
Accepted
```

Q3. Regex Text Search Engine

CODE:

```
import re
text = """Meeting on 12/09/2026
Call 9876543210
#NLP
@OpenAI
natural language processing"""
print("1. Word Search")
print("2. Prefix Search")
print("3. Suffix Search")
print("4. Date Search")
print("5. Phone Number Search")
print("6. Hashtag Search")
print("7. Mention Search")
choice = input("Enter choice: ")
if choice == "1":
    word = input("Enter word: ")
    result = re.findall(r"\b" + re.escape(word) + r"\b", text, re.I)
elif choice == "2":
    word = input("Enter prefix: ")
    result = re.findall(r"\b" + re.escape(word) + r"w*", text, re.I)
elif choice == "3":
    word = input("Enter suffix: ")
    result = re.findall(r"w*" + re.escape(word) + r"\b", text, re.I)
elif choice == "4":
    result = re.findall(r"\b\d{2}/\d{2}/\d{4}\b", text)
elif choice == "5":
    result = re.findall(r"\b[6-9]\d{9}\b", text)
elif choice == "6":
    result = re.findall(r"#\w+", text)
elif choice == "7":
    result = re.findall(r"@w+", text)
else:
    result = []
print("\nMatching Patterns:", result)
print("Total Matches:", len(result))
```

main.py

Visualize

Run ▶

Output

```
1 import re
2 text = """Meeting on 12/09/2026
3 Call 9876543210
4 #NLP
5 @OpenAI
6 natural language processing"""
7 print("1. Search Date")
8 print("2. Search Phone Number")
9 print("3. Search Hashtag")
10 print("4. Search Mention")
11 print("5. Search Prefix")
12 print("6. Search Suffix")
13 print("7. Search Word")
14 choice = input("Enter choice: ")
15 if choice == "1":
16     result = re.findall(r"\b\d{2}/\d{2}/\d{4}\b", text)
17 elif choice == "2":
18     result = re.findall(r"\b[6-9]\d{9}\b", text)
19 elif choice == "3":
20     result = re.findall(r"#\w+", text)
21 elif choice == "4":
22     result = re.findall(r"@w+", text)
23 elif choice == "5":
24     word = input("Enter prefix: ")
25     result = re.findall(r"\b" + re.escape(word) + r"\w*", text, re.I)
26 elif choice == "6":
27     word = input("Enter suffix: ")
28     result = re.findall(r"\w*" + re.escape(word) + r"\b", text, re.I)
29 elif choice == "7":
30     word = input("Enter word: ")
31     result = re.findall(r"\b" + re.escape(word) + r"\b", text, re.I)
32 else:
33     result = []
34 print("\nMatching Patterns:", result)
35 print("Total Matches:", len(result))
```

1. Search Date
2. Search Phone Number
3. Search Hashtag
4. Search Mention
5. Search Prefix
6. Search Suffix
7. Search Word
Enter choice: 2

Matching Patterns: ['9876543210']
Total Matches: 1